

Introduction to Probability

Not-so-bad stuff

Combinatorics, DRVs (Bernoulli, Binomial, Poisson, Geometric, Negative Binomial, Hypergeometric), CRVs (Uniform, Exponential, Normal), expectation, moments, variance.

Marginal and Joint Distributions

Independence, Covariance, Correlation

Limit Theorems and Inequalities

Tailedness Inequalities

Name	Conditions	Formula
Markov	RV $X \geq 0$, $\mathbf{E}[X] \in \mathbb{R}$	$P[X \geq a] \leq \frac{\mathbf{E}[X]}{a}$
Chebyshev	$\mathbf{E}[X], \mathbf{V}[X] \in \mathbb{R}$	$P[\ X - \mathbf{E}[X]\ \geq a] \leq \frac{\mathbf{V}[X]}{a^2}$

Limit Theorems

Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ where X_i are i.i.d. RVs.

Name	What	Theorem
Weak Law of Large Numbers (WLoN*)	Converges to	$\lim_{n \rightarrow \infty} \mathbf{P}[\ \bar{X}_n - \mu\ > \varepsilon] = 0$
Strong Law of Large Numbers (SLoN*)	Converges to (non-exam.)	$\mathbf{P}\left[\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right] = 1$
Central Limit Theorem (CLT)	How it converges	$\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma} \sim N(0, 1)$

*Only I call it that

**Technically $\lim_{n \rightarrow \infty} F_{Z_n}(a) = \Phi(a)$